



Dhawal Salwan

Doctor of Philosophy in Electrical Engineering

Specialization: Wireless Communication

Indian Institute of Technology Ropar

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EDUCATION

Degree	Institute/Board	CGPA/Percentage	Year
Ph.D. in Electrical Engineering	Indian Institute of Technology Ropar, India	8.00	2021-2026
Bachelor of Engineering (ECE)	Thapar Institute of Engineering and Technology, Patiala, India	8.33	2017-2021
Senior School Certificate Examination	Central Board of Secondary Education, New Delhi	77.8%	2016
Secondary School Examination	Central Board of Secondary Education, New Delhi	8.2	2014

RESEARCH INTERESTS

My research interests lie in the broad area of wireless communications, with a particular focus on joint sensing and communication, UAV networks, and advanced air mobility.

PUBLICATIONS

Journals

- **D. Salwan**, S. Agarwal, and B. Kumbhani, "Design and Analysis of a Low-Complexity Communication Receiver for PSK-LFM Joint Sensing and Communication Waveform," **IEEE Transactions on Intelligent Vehicles**, 2025. DOI: 10.1109/TIV.2025.3621906.
- **D. Salwan**, S. Agarwal, and B. Kumbhani, "Practicalities in Communication Receiver Design for JSC Waveforms: Whether to Filter or to Undersample?," **IEEE Communications Letters**, 2025. DOI: 10.1109/LCOMM.2025.3633892.

Conferences

- **D. Salwan**, S. Agarwal, B. Kumbhani, S. Darshi, K. Namuduri, S. Chakravarty, and A. Ashok, "Communication Receiver Design for PSK-LFM Joint Sensing and Communication Waveform," **Proceedings of the 2024 IEEE International Conference on Communications Workshops (ICC Workshops)**, Denver, CO, USA, 2024, pp. 1617–1622.
- N. Gupta, A. Ahmad, **D. Salwan**, S. Agarwal, and K. Namuduri, "Revolutionizing Air Mobility: Cutting-Edge Infrastructure for Safe Air Corridor Navigation," **IEEE Future Networks World Forum (FNWF)**, Bangalore, India, 2025.

FIRST-AUTHOR UNDER REVIEW PUBLICATIONS

1. IEEE Internet of Things Journal.
2. IEEE Transactions on Aerospace and Electronic Systems.
3. IEEE International Conference on Advanced Networks and Telecommunications Systems (ANTS).

AWARDS / ACHIEVEMENTS

- Received First Prize in the IEEE Future Networks World Forum (IEEE FNWF) Student Tech Poster Competition, along with a \$1,000 travel grant to attend flagship IEEE ComSoc conferences, including IEEE GLOBECOM 2025.
- Received the IEEE ComSoc Travel Grant of \$1,800 for attending and presenting a paper at the IEEE International Conference on Communications (IEEE ICC), 9–13 June 2024.
- Awarded 3rd Prize in the IEEE VTS Innovation Challenge on Advanced Air Mobility, 2022.
- Qualified GATE 2021 with an All India Rank of 3031, and GATE 2025 with an All India Rank of 6287.
- Selected for the Direct Ph.D. program in the Department of Electrical Engineering at IIT Ropar, directly after completing my Bachelor of Engineering.
- Received a participation certificate for attending a 3-day workshop on "5G and Beyond – What Next," organized by IIT Kanpur, 12–14 December 2022.
- Attended the Indian Mobile Congress (IMC) 2024 at Pragati Maidan, New Delhi, which showcased advancements in 5G and beyond, including innovations from IITs, academia, and leading industries.
- Presented research posters during Research Scholar Day 2024 and 2025 at IIT Ropar, showcasing ongoing research on Joint Sensing and Communication.
- Received a participation certificate for attending the workshop on Intelligent Transportation Systems and IoT, organized by the Department of Electrical Engineering, IIT Ropar, in 2024.
- Secured A grades in Signals and Systems, Physics, Manganal Project, Modern Control Theory, Graph Theory, Biomedical Signal Processing, Capstone Project, and Fibre Optic Communication during B.E. at Thapar University.
- Awarded a participation certificate for designing and demonstrating a working FM Radio Transmitter during B.E. in 2019.

TECHNICAL SKILLS

- **Programming:** C, C++, Python
- **Simulation & Modeling:** MATLAB for wireless communication system simulation
- **Documentation:** L^AT_EX for academic writing

COURSES TAKEN DURING PH.D.

- Advanced Digital Communication (**Grade: A-**)
- Analytical Techniques for Signal Processing & Communication (**Grade: B**)
- Wireless Ad-hoc Networks (**Grade: B**)
- Detection & Estimation Theory (**Grade: B**)
- Adaptive Signal Processing (**Grade: B**)
- Special Topics in Communication (**Grade: B-**)
- Optimization Techniques (**Grade: C**)
- Modern Radar Systems (**Grade: A**)

EXPERIENCE

Teaching / Research Assistant Ropar, India	<i>IIT Ropar</i> <i>Aug. 2021 – May 2026</i>
• Assisted in conducting laboratory classes for Basic Electrical Engineering, Basic Electronics Engineering, and Advanced Digital Communication Laboratory, including hands-on experiments using Software Defined Radio (SDR). • Assisted in designing and evaluating mid-semester and end-semester exams for Signals and Systems and Advanced Digital Communication.	

POSITIONS OF RESPONSIBILITY

• Mentor Representative , Department of ECE, Thapar University, Patiala	<i>Aug. 2018 - May. 2021</i>
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